



SCHOOL OF
COMPUTING

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Exercise-6: PCA

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CLASS : CSE-B

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2. For the below dataset, identify which variable is principal component.

Variable	1	2	3
Climate	0.190	0.017	0.207
Housing	0.544	0.020	0.204
Health	0.782	-0.605	0.144
Crime	0.365	0.294	0.585
Transportation	0.585	0.085	0.234
Education	0.394	-0.273	0.027
Arts	0.985	0.126	-0.111
Recreation	0.520	0.402	0.519
Economy	0.142	0.150	0.239

```
In [5]: import pandas as pd
import numpy as np

data = {
    'variable': [
        'Climate',
        'Housing',
        'Health',
        'Crime',
        'Transportation',
        'Education',
        'Arts',
        'Recreation',
        'Economy'
    ],

    'PC1': [
        0.190,
        0.544,
        0.782,
        0.365,
        0.585,
        0.394,
        0.985,
        0.520,
        0.142
    ],

    'PC2': [
        0.017,
        0.020,
        -0.605,
        0.294,
        0.085,
        -0.273,
        0.126,
        0.402,
```

```

        0.150
    ],
    'PC3': [
        0.207,
        0.204,
        0.144,
        0.585,
        0.234,
        0.027,
        -0.111,
        0.519,
        0.239
    ]
}

df = pd.DataFrame(data)

print("Principal Component Dataset")
print("=" * 60)
print(df.to_string(index=False))

```

Principal Component Dataset

```

=====
Variable  PC1    PC2    PC3
Climate  0.190  0.017  0.207
Housing  0.544  0.020  0.204
Health   0.782 -0.605  0.144
Crime    0.365  0.294  0.585
Transportation 0.585  0.085  0.234
Education 0.394 -0.273  0.027
Arts     0.985  0.126 -0.111
Recreation 0.520  0.402  0.519
Economy  0.142  0.150  0.239

```

```
In [6]: for pc in ['PC1', 'PC2', 'PC3']:

        df['Absolute_' + pc] = df[pc].abs()

        index = df['Absolute_' + pc].idxmax()

        variable = df.loc[index, 'Variable']
        value = df.loc[index, pc]

        print("\n", pc)
        print(" - " * 30)
        print("Principal Variable :", variable)
        print("Loading           :", value)
        print("Absolute Loading    :", abs(value))
```

PC1

```
-----
Principal Variable : Arts
Loading           : 0.985
Absolute Loading  : 0.985
```

PC2

```
-----
Principal Variable : Health
Loading           : -0.605
Absolute Loading  : 0.605
```

PC3

```
-----
Principal Variable : Crime
Loading           : 0.585
Absolute Loading  : 0.585
```

```
In [7]: for pc in ['PC1', 'PC2', 'PC3']:

        print("\n" + "=" * 60)
        print("VARIABLE CONTRIBUTIONS FOR", pc)
        print("=" * 60)

        result = df[['Variable', pc]].copy()

        result['Absolute Loading'] = result[pc].abs()

        result = result.sort_values(
            by='Absolute Loading',
            ascending=False
        )

        print(result.to_string(index=False))
```

```
=====
VARIABLE CONTRIBUTIONS FOR PC1
=====
```

Variable	PC1	Absolute Loading
Arts	0.985	0.985
Health	0.782	0.782
Transportation	0.585	0.585
Housing	0.544	0.544
Recreation	0.520	0.520
Education	0.394	0.394
Crime	0.365	0.365
Climate	0.190	0.190
Economy	0.142	0.142

```
=====
VARIABLE CONTRIBUTIONS FOR PC2
=====
```

Variable	PC2	Absolute Loading
Health	-0.605	0.605
Recreation	0.402	0.402
Crime	0.294	0.294
Education	-0.273	0.273
Economy	0.150	0.150
Arts	0.126	0.126
Transportation	0.085	0.085
Housing	0.020	0.020
Climate	0.017	0.017

```
=====
VARIABLE CONTRIBUTIONS FOR PC3
=====
```

Variable	PC3	Absolute Loading
Crime	0.585	0.585
Recreation	0.519	0.519
Economy	0.239	0.239
Transportation	0.234	0.234

Climate	0.207	0.207
Housing	0.204	0.204
Health	0.144	0.144
Arts	-0.111	0.111
Education	0.027	0.027

```
In [8]: import matplotlib.pyplot as plt

variables = df['variable']

x = np.arange(len(variables))
width = 0.25

plt.figure(figsize=(12, 7))

plt.bar(x - width, df['PC1'], width, label='PC1')
plt.bar(x, df['PC2'], width, label='PC2')
plt.bar(x + width, df['PC3'], width, label='PC3')

plt.xticks(x, variables, rotation=45)
plt.xlabel("Variables")
plt.ylabel("Loading Value")
plt.title("Principal Component Loadings")
plt.legend()
plt.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.show()
```

